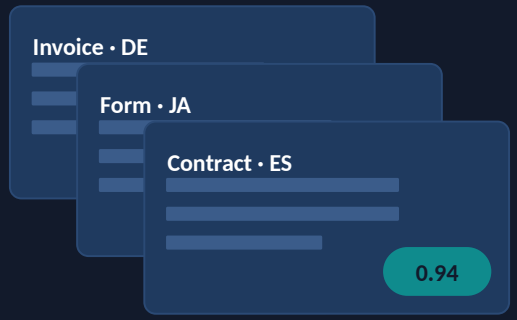
DATANOVA 2026 GLOBAL DATA-THON · TRACK DN-AI-01



Lingua Ledger

Self-verifying extraction for multilingual business documents. Every field comes back with a confidence score you can actually trust.

Rishav Naskar · rishavnaskar.r@gmail.com · Round 1 submission, November 2026

Every language and layout arrives. Trust does not

Official challenge DN-AI-01: extract, classify and structure information from multilingual invoices, contracts and forms with varying layouts and quality, and output structured, validated data with confidence scores.

Multilingual by default

Finance, legal and operations teams in India, the EU and South-East Asia process invoices and contracts in 5–10 languages, often mixed on a single page.

No two layouts alike

Template-based OCR breaks on every new vendor. LLMs read anything, but they also confidently invent a VAT number when the scan is blurry.

100% human review, or silent errors

Because nobody can tell which fields are right, teams either re-key everything or accept errors into the ledger. Both are expensive.

A confidence score that actually means something

Approach	Layout robustness	Confidence	Human review
Template OCR	Brittle; fails on new layouts	Word-level OCR confidence only	Every exception to a human
Prompt-and-pray LLM	Reads anything	Self-reported, uncalibrated, often wrong	All or nothing
Lingua Ledger	Vision LLM + OCR LLM, cross-checked	Calibrated per field: 0.9 means ~90% correct	Only low-confidence fields

Our headline metric: at a target field accuracy (e.g. 98%), what share of fields still needs a human? Lower is better, and it is measurable.

Read twice, validate, calibrate, route

Ingest and classify

PDF, photo or scan in. Pages rendered, language and document type detected (invoice, receipt, form, contract).

Extract twice, independently

Path A: a vision model reads the page against a strict JSON schema. Path B: multilingual OCR text + layout, same schema, text-only model.

Validate deterministically

Line items must sum, tax maths must hold, dates must parse and order, IBAN / VAT / currency formats must match. Every rule is explainable.

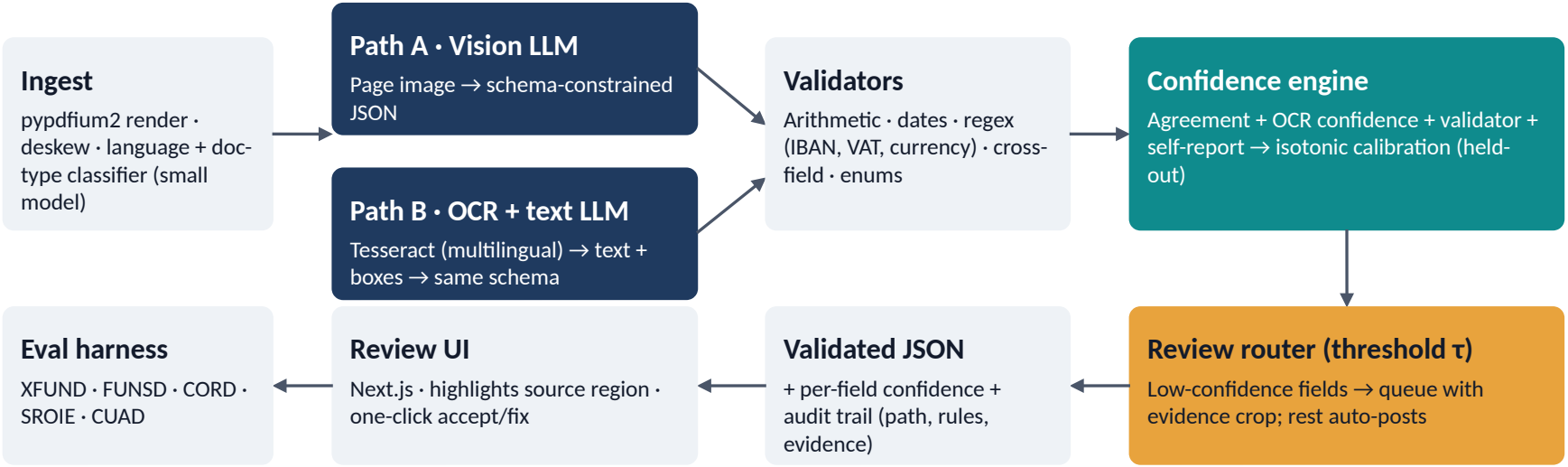
Calibrate confidence per field

Path agreement, OCR word confidence under the field, validator outcome and model self-report are fused, then isotonic-calibrated on held-out data.

Route only what needs eyes

Fields under the threshold go to a review queue with the source region highlighted. Everything else posts automatically, with an audit trail.

Two independent readers, one deterministic judge



Stack: Python 3.13, pydantic schemas, Google Gemini (gemini-3.5-flash-lite for vision, text and classification, structured JSON output; provider switchable to Claude), Tesseract, scikit-learn (isotonic regression), FastAPI, Next.js on Vercel. Public repo, MIT licence, one-command reproduction of every metric.

What makes 0.9 mean ninety percent

SIGNALS PER FIELD

Path agreement
Do the vision reader and the OCR reader return the same normalised value?

OCR evidence
Mean Tesseract word confidence inside the field's bounding box; low on blur, skew, stamps.

Validator outcome
Passed, failed or not applicable for each deterministic rule touching the field.

Model self-report
The LLM's own 0–1 estimate, used as a weak feature, never as the answer.

CALIBRATION AND ROUTING

- Signals are fused by a small model, then isotonic-regressed against field correctness on a held-out split, per document type.
- Reported on a reliability diagram with Expected Calibration Error (ECE) before and after calibration.
- A single threshold τ turns confidence into a routing decision; we publish the full review-rate versus accuracy curve so an operator picks their own point.
- Evidence travels with every field: which path, which rules fired, the crop of the page it came from.

Public benchmarks, scored per language and per field

Dataset	Document type	Languages	What we measure
XFUND v1.0 (Microsoft)	Forms	ZH, JA, ES, FR, IT, DE, PT	Key–value extraction F1 per language
FUNSD	Forms	EN	Key–value F1; English baseline
CORD v2 (Naver Clova)	Receipts	ID (Latin script)	Field F1; arithmetic validator hit-rate
SROIE (ICDAR 2019)	Receipts / invoices	EN	Exact-match F1 on company, date, address, total
CUAD (Atticus Project)	Contracts	EN	Clause extraction F1 on parties, dates, governing law, termination



Ablation

Vision-only vs OCR-only vs dual-path + validators, same prompts, same schema.



Calibration

Reliability diagram, ECE, Brier score; before vs after isotonic fit, per doc type.



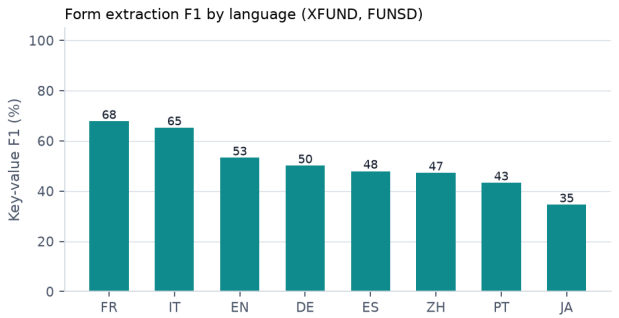
Cost and latency

p50 / p95 seconds per page and dollars per 1,000 documents, by path.

All datasets are public research benchmarks used under their published licences; no customer documents are used. Synthetic multilingual invoices rendered from templates cover the invoice type across the same seven languages.

Results on 198 public benchmark documents

Benchmark	Vision only	OCR + text only	Both + validators
XFUND forms (7 langs)	51%	32%	**51%**
FUNSD forms (EN)	54%	39%	**53%**
CORD receipts (ID)	95%	52%	**95%**
SROIE receipts (EN)	95%	89%	**95%**
Synthetic invoices (7 langs)	99%	96%	**99%**
CUAD contracts (EN)	61%	60%	**60%**



92%

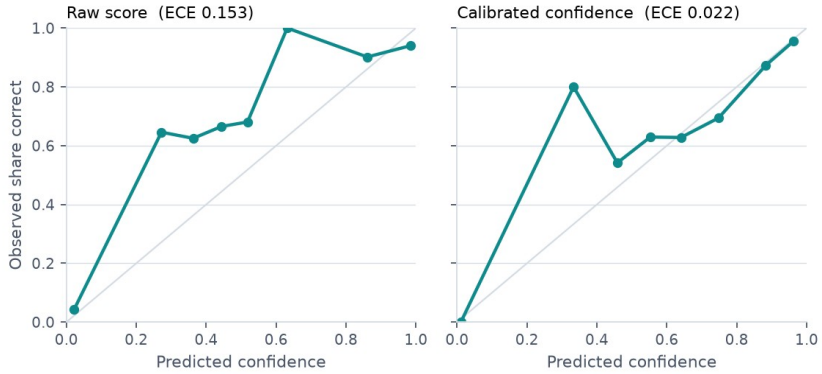
document type detected correctly

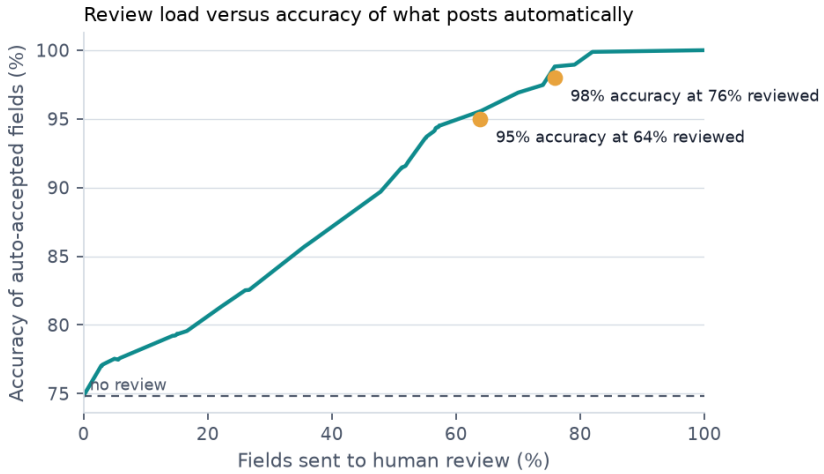
89%

language detected correctly

Field-level F1 against published ground truth. Forms: key-value pairs. Receipts and invoices: every labelled field and line item. Contracts: parties, dates, governing law. Same prompts and schema for every path.

Calibrated confidence turns into a review budget





0.153 → 0.022

expected calibration error, raw score → calibrated (lower is better)

75%

fields correct with no human review at all

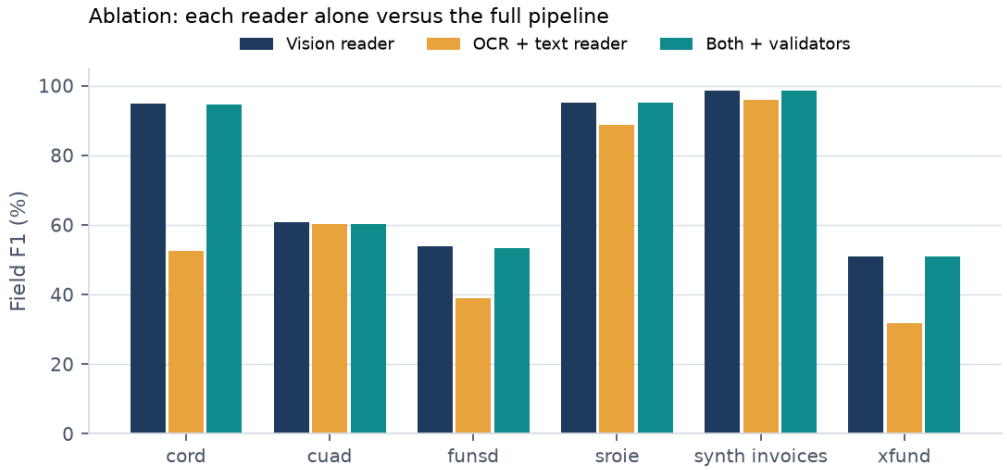
76%

of fields need a human to reach 98% accuracy on what posts automatically

64%

of fields need a human to reach 95% accuracy

What it costs to run



Tokens per document: about 2435 in, 1548 out across both readers. Review effort scales with low-confidence fields, not with document volume.

11.8 s

median model reading time per document, both readers in parallel, Free Tier rate limits included

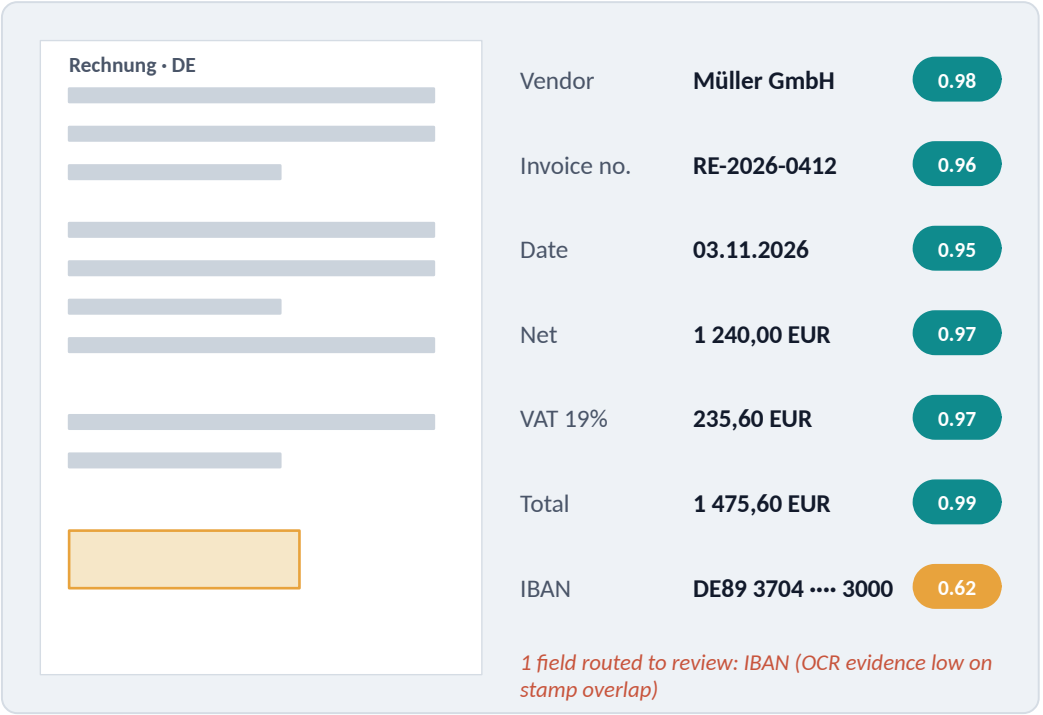
21.2 s

median end to end, with OCR, classification and throttling waits

$4.60

model cost per 1,000 documents at list prices

Demo: drop a document, watch it verify itself



- Upload any PDF or photo; sample documents in seven languages provided.
- Side-by-side view of what each reader saw and where they disagreed.
- Validator panel: which rules fired, which passed, in plain words.
- Threshold slider: move τ and watch review load and expected accuracy change live.
- Export validated JSON or CSV with the audit trail attached.

Impact: fewer eyes per document, no silent errors

Who benefits

Accounts-payable and procurement teams, legal operations, and BPOs serving multilingual markets (India, EU, South-East Asia, LATAM).

Economics

Review effort scales with low-confidence fields, not with document volume. Cost per document is dominated by one or two model calls per page and is reported in the evaluation.

Scalability

Stateless pipeline; each stage is independently parallel. New document types are a schema and a validator file, not a retrained model.

Responsible by design

- Human stays in the loop exactly where the system is unsure; no autonomous posting below τ.
- Every field carries its evidence, so decisions are auditable after the fact.
- Documents are processed in memory and never used for training; PII fields can be masked in exports.
- Calibration is reported per language and per document type to expose, not hide, weaker segments.
- Public benchmarks, public code, public numbers.

Roadmap: pipeline, then evidence, then production

1

Round 1 · 10 Nov

Working pipeline with evidence

- Runs both readers, validators, calibration, and review routing
- 198 public benchmark documents, 9 languages; one command, all numbers
- Live hosted demo with sample documents
- 225 unit tests under an MIT licence

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Round 2 · by 25 Nov

Better readers, same pipeline

- Bigger Gemini, Claude readers; one environment variable; compared cost, accuracy
- Japanese and Chinese forms: layout-aware prompts, key normalisation
- Lenient form scoring right next to the strict one
- Corrections from the review UI feed the calibrator

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Beyond Round 2

Moving into production

- Active learning straight from reviewer corrections
- Open-weights model path for on-prem use
- Connectors for email inbox, SAP, and Tally
- Full table and handwriting extraction support

TEAM AND CONTRIBUTIONS

RN

Rishav Naskar

Software engineer · AI agents, LLM applications and full-stack web · Solo entrant

- Problem framing, architecture and schema design
- Dual-path extraction pipeline, validators and confidence engine
- Evaluation harness, calibration and reporting
- Review UI, hosted demo, documentation and video

Third-party components acknowledged in the repository: XFUND, FUNSD, CORD, SROIE and CUAD datasets; Tesseract OCR; Google Gemini API; pypdfium2; scikit-learn; FastAPI; Next.js. All code released under the MIT licence.

Lingua Ledger · rishavnaskar.r@gmail.com · repo and demo links in the submission form